

Even Term (Even) Semester Regular/Back Examination May 2026

Roll no. 2319388

Name of the Program and Semester: B. Tech (CSE) AI-ML, 6th Semester

Name of the Course: Transformer Model and Applications

Course Code: TCS-666

Time: 3 hours

Maximum Marks: 100

Note:

- All Questions are compulsory
- Answer any two sub questions among a, b and c in each main question.
- Total marks in each main question are twenty
- Each question carries 10 marks.

Q1.

(10×2=20 Marks) CO1

- a. Describe Masked attention:

$$S = \frac{QK^T}{\sqrt{d_k}} + M$$

1. Prove masked positions get zero probability after softmax.
2. Show that masking enforces causal structure mathematically.
3. Derive effect on gradient flow.
4. Explain its necessity in autoregressive decoding.

- b. Consider the sinusoidal positional encoding defined as:

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right), PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

Prove that for any fixed offset k , the positional encoding at position $pos+k$ can be represented as a linear function of the encoding at position pos . What implication does this have for attention mechanisms?

- c. Explain The feed-forward network (FFN) in Transformers is position-wise:

$$FFN(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

Why is FFN applied independently to each position? What role does it play despite attention already mixing information?

Q2.

(10×2=20 Marks) CO2

- a. Consider a The Transformer uses the Adam optimizer with learning rate warmup and decay. Explain the learning rate schedule:

$$lr = d_{model}^{-0.5} \cdot \min(step^{-0.5}, step \cdot warmup^{-1.5})$$

Discuss:

1. Why warmup is necessary in early training stages?
2. How this schedule stabilizes training compared to constant learning rate
3. What would happen if warmup is removed?



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- b. Explain different batch processing techniques such as fixed size batching, dynamic batching, padding and masking also analyze how does improper padding affect computational efficiency?
- c. What is meant by convergence during the training of a Transformer model? How can you identify whether a model has converged?

Q3.

(10×2=20 Marks) CO3

- a. Explain the process of text generation using GPT (Generative Pre-trained Transformer) models such as GPT models with suitable code.
- b. Explain BERT (Bidirectional Encoder Representations from Transformers) and its variants. Discuss how BERT is used for different NLP tasks and compare it with its major variants.
- c. Compare Machine Translation and Text Summarization in the context of Transformer-based NLP models.
Discuss:
 - 1. Similarities in architecture
 - 2. Differences in objectives
 - 3. Training data requirements
 - 4. Evaluation techniques
 - 5. Challenges in real-world applications

Q4.

(10×2=20 Marks) CO4

- a. Explain how an image is converted into a sequence of tokens in the Vision Transformer. How does this process make images suitable for Transformer architecture?
- b. Describe the architecture of a Transformer used in audio processing tasks. How is it similar to and different from text-based Transformers?
- c. Explain the concept of cross-modal learning. How does it enable models to perform joint tasks using multiple data modalities?

Q5.

(10×2=20 Marks) CO5

- a. Discuss the impact of sequence length on the computational complexity of Transformers. Why does it create scalability challenges?
- b. Discuss What is bias in datasets? How can biased data affect the performance and fairness of machine learning models?